CS23331-Design and Analysis of Algorithms

Problem 1: Finding Complexity using Counter Method

Completion requirements

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
    int s =1;
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and printf() statements. count variable

Input:
A positive Integer n

Output:
Print the value of the counter variable

For example:

Input	Result
9	12

Answer:

```
#include<stdio.h>

int main(){

   int n;

   scanf("%d",&n);

   int i=1;

   int s=1;

   int count=2;

   while(s<=n){
```

```
    count++;
    i++;
    count++;
    s+=i;
    count++;
  }
  int answer=count+1;
  printf("%d",answer);
  return 0;
}
```

Input	Expected	Got	
	9	12	12
	4	9	9

Problem 2: Finding Complexity using Counter method

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
```

```
      }
    }
  }
}
```

Note: No need of counter increment for declarations and scanf() and printf() statements. count variable

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>

int main(){

   int n;

   scanf("%d",&n);

   int count=0;

   count++;

   if(n==1)

   {

    printf("*");

   }

   else

   {

    for(int i=1; i<=n; i++)

    {

     count++;

     for(int j=1; j<=n; j++)

     {

       count++;

       count++;

       count++;

       break;

     }

     count++;;
```

```
    }
     count++;
   }
   printf("%d",count);
   return 0;
}
```

Input	Expected	Got	
	2	12	12	
	1000	5002	5002	
	143	717	717	

Problem 3: Finding Complexity using Counter Method

Convert the following algorithm into a program and find its time complexity using counter method.

```
 Factor(num) {
 {
    for (i = 1; i <= num;++i)
    {
     if (num % i== 0)
        {
          printf("%d ", i);
        }
     }
  }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:
A positive Integer n
Output:
Print the value of the counter variable

Answer:

```
#include <stdio.h>


int main()
```

```
{
   int num;
   scanf("%d",&num);
   int count = 0;
   int i;
   for(i = 1; i <= num; i++)
   {
      count++;

      if(num % i == 0)
      {
         count++;
      }
      count++;
   }
   count++;
   printf("%d",count);

   return 0;
}
```

Input	Expected	Got	
	12	31	31
	25	54	54
	4	12	12

Problem 4: Finding Complexity using Counter Method

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and printf() statements. count variable

Input:
A positive Integer n
Output:
Print the value of the counter variable

Answer:

```
#include<stdio.h>

int main(){

   int n;

   scanf("%d",&n);

   int c=0;

   int count=2;

   for(int i=n/2;i<n;i++){

      count++;

      for(int j=1;j<n;j=2*j){

         count++;

         for(int k=1;k<n;k=k*2){

            count++;

            c++;

            count++;

         }

         count++;

      }

      count++;

   }

   printf("%d",count);
```

```
   return 0;
}
```

Input	Expected	Got	
	4	30	30
	10	212	212

Problem 5: Finding Complexity using counter method

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and printf() statements. count variable

Input:
A positive Integer n
Output:
Print the value of the counter variable

Answer:

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int rev=0,rem;
    int count=0;
    while(n!=0){
        count++;
        rem=n%10;
        count++;
        rev=rev*rem;
        count++;
        n=n/10;
        count++;
```

```
    }
    count+=3;
    printf("%d",count);
    return 0;
}
```

Input	Expected	Got	
	12	11	11
	1234	19	19